

Web Technologies - HTML Assignment

Q1. HTTP Request–Response Cycle

The HTTP Request–Response Cycle is the process through which a client (browser) communicates with a web server to get a resource such as a webpage.

Major Steps:

- The user enters a URL in the browser.
- The browser sends an HTTP request to the web server.
- The web server receives and processes the request.
- The server sends an HTTP response back to the browser.
- The browser receives the response and displays the webpage.

Diagram:

Client (Browser) → HTTP Request → Web Server

Client (Browser) ← HTTP Response ← Web Server

Q2. Introduction to Web Technologies

1. HTML

HTML stands for HyperText Markup Language. It is used to create the structure and content of a webpage.

2. CSS

CSS stands for Cascading Style Sheets. It is used to style webpages by adding colors, fonts, spacing, layouts, and designs.

3. JavaScript

JavaScript is a programming language used to make webpages interactive and dynamic.

4. Bootstrap

Bootstrap is a CSS framework used to create responsive and mobile-friendly websites quickly.

5. jQuery

jQuery is a JavaScript library that makes DOM manipulation, events, and animations easier.

6. React.js

React.js is a JavaScript library used to build interactive and reusable user interfaces.

Q3. Client-Side vs. Server-Side Technologies

Client-side technologies run inside the user's web browser. Examples include HTML, CSS, and JavaScript.

Server-side technologies run on the web server and process requests before sending a response to the client. Examples include Java, Python, PHP, and Node.js.

Client-Side	Server-Side
Runs in the browser	Runs on the server
Used for user interface and interaction	Used for processing and database operations
Examples: HTML, CSS, JavaScript	Examples: Java, Python, PHP, Node.js

Q4. Block-Level Elements in HTML

Block-level elements are HTML elements that normally start on a new line and take up the full available width.

Characteristics:

- They start on a new line.
- They usually take the full width available.
- They can contain other elements.

Examples:

- <div>
- <p>
- <h1> to <h6>
- <section>

Q5. Semantic HTML Elements

Semantic HTML elements clearly describe the meaning and purpose of the content they contain.

They make webpages easier to understand, improve accessibility, and help search engines understand the webpage structure.

Examples:

- <header> - Defines the header of a page.
- <nav> - Defines navigation links.
- <section> - Defines a section of content.
- <article> - Defines independent content.
- <footer> - Defines the footer of a page.

Q6. Creating a Table Using HTML

Sr. No.	Name	Designation	Department
1	John	Manager	Sales
2	Smith	Sr. Manager	Marketing
3	Jane	Manager	HR

Q7. Creating a Nested List Using HTML

- River Ganga
- Yamuna
- Brahmaputra
- Narmada

Q8. Types of CSS

1. Inline CSS

Inline CSS is written directly inside an HTML element using the style attribute.

This is an example of Inline CSS.

2. Internal CSS

Internal CSS is written inside the <style> tag in the <head> section of an HTML document.

3. External CSS

External CSS is written in a separate .css file and connected to the HTML file using the <link> tag.

Example: <link rel="stylesheet" href="style.css">

Q9. Practical Understanding

In simple words: HTML = Structure, CSS = Style, JavaScript = Behavior.

Technology	Role	What it contributes
HTML	Structure	Creates headings, paragraphs, images, tables, etc.
CSS	Design	Adds colors, fonts, spacing, layouts, and styling.
JavaScript	Behavior	Adds interaction and dynamic features.

Q10. Short Answer

a. What is the role of Bootstrap?

Bootstrap is a CSS framework used to create responsive, attractive, and mobile-friendly websites easily.

b. What is jQuery?

jQuery is a JavaScript library that simplifies HTML DOM manipulation, event handling, animations, and other common JavaScript tasks.

c. What is React.js?

React.js is a JavaScript library used to create interactive user interfaces. It helps developers build websites using reusable components and efficiently update the webpage.